

Statement of Work

For

Air Force Institute of Technology (AFIT) Engineering Department (ENG) Cleanroom AQ6380-10-L1-FCC/Z 900-1650 Highest Performance Optical Spectrum Analyzer (OSA)

27 February 2025

I. Background:

The Air Force Institute of Technology (AFIT) Electrical & Computer Engineering Department (ENG) Cleanroom processes silicon wafers for production of Microelectromechanical (MEMS) and Nano electronic circuits for Cleanroom courses and student research projects for thesis presentation in the graduate school of Engineering. The current Yokogawa AQ6370C used to measure optical spectra from 900 to 1650nm of fabricated fabricate integrated MEMS circuitry optical process device. The AFIT ENG Cleanroom is in area B building B644 of Wright-Patterson Air Force Base in Dayton, OH. The current AFIT/ENG owned AQ6370C Optical Spectrum Analyzer (OSA) is 20 years old and no longer meets the needs for measuring optical spectra wavelength ranges of 900 to 1650 nm.

“New” replacement AQ6380-10-L1-FCC/Z 900-1650 offers the highest performance OSA, which is an indispensable and essential tool used to support Cleanroom courses EENG 636, EENG 699, EENG 777, EENG 799, EENG 899, and EENG 999 courses, designed to provide precise and reliable measurement of optical spectra within the 900 to 1650 nm wavelength range. The Yokogawa AQ6380 OSA will be better prepare students to contribute to advancements in optical technology for the Air and Space Forces, enhancing their technical skills and understanding of key concepts in optical engineering.

II. Minimum Technical Requirements:

a. The following specification lists the minimum requirements for the AQ6380-10-L1-FCC/Z 900-1650 - Highest Performance Optical Spectrum Analyzer HARD CARRING CASE AQ6370 - A1114AF-G

AQ6380 delivers industry-leading resolution as fine as 5 pm (approximately 0.624 GHz). Level of detail enables more accurate characterization of dense spectral lines and frequency combs, providing critical clarity in cutting-edge photonics research and product development. The 5 pm resolution is four times better than traditional models offering 20 pm (2.5 GHz) resolution, ensuring measurements capture even the narrowest spectral features. AQ6380 covers numerous optical communication bands

(including O, E, S, C, L) and emerging photonics applications. The broad wavelength coverage is 900–1650 nm. The broad range reduces the need for multiple instruments, streamlining testing and saving both time and resources.

AQ6380 offers multiple sensitivity modes, achieving levels down to –85 dBm in the 1200–1600 nm range. Its advanced sensitivity ensures accurate detection of low-power signals without sacrificing speed or measurement accuracy. AQ6380 maintains a high dynamic range for clean, noise-free results. The dynamic range avoids the overlap or saturation often encountered in lesser OSAs, producing more reliable data when measuring multiple lines in a spectrum. AQ6380 provides rapid scan speeds, boosting throughput in production environments and accelerating research timelines and minimizes downtime and testing bottlenecks. A clear, intuitive interface coupled with built-in data analysis functions simplifies complex measurements. AQ6380 has fewer manual steps and built-in peak analysis routines help novice operators and expert users alike gather and interpret results efficiently. Yokogawa is globally recognized for precision test equipment. Their robust design, customer support, and long-term maintenance options ensure minimal operational interruptions and a lower total cost of ownership. The AQ6380's stable performance over years of intensive use further bolsters Yokogawa's reputation as a dependable solution provider.

Terms and Conditions: Lead Time: 10-12 Weeks ARO

Payment: Net 30

III. Warranty: Three (3) year warranty

IV. Training: Not required. The Optical Analyzer is the model replacement of current outdated, obsolete Yokogawa Optical Analyzer.

V. Installation: Not required. Cleanroom Technician will setup the Analyzer.

VI. Delivery:
SHIP TO DoDAAC: F4F5AN
POC: Mr. Michael Hill Sr. (POC)
AFIT/ENG Bldg.644 Rm320
2950 Hobson Way
Wright-Patterson AFB, OH

VII. Lead Time:
The contractor shall deliver ordered systems 90 days after contract approval.

VIII. Period of Performance

March 10, 2025 – July 24, 2025

- IX.** **Delivery Procedures** – Shipped by freight through base supply to AFIT/ENG mailroom.